

What is an Oracle Triggers?

- A trigger is a named PL/SQL block stored in the Oracle Database and executed automatically when a triggering event takes place. The event can be any of the following:
- A data manipulation language (DML) statement executed against a table e.g., INSERT, UPDATE, or DELETE.
- For example, if you define a trigger that fires before an INSERT statement on the customers table, the trigger will fire once before a new row is inserted into the customers table.
- A data definition language (DDL) statement executes e.g., CREATE or ALTER statement. These triggers are often used for auditing purposes to record changes of the schema.
- A system event such as startup or shutdown of the Oracle Database.
- A user event such as login or logout.
- The act of executing a trigger is also known as firing a trigger. We say that the trigger is fired.

Oracle Trigger Usage

Oracle triggers are useful in many cases such as the following:

- Enforcing complex business rules that cannot be established using integrity constraint such as UNIQUE, NOT NULL, and CHECK.
- Preventing invalid transactions.
- Gathering statistical information on table accesses.
- Generating value automatically for derived columns.
- Auditing sensitive data.

Question #1 of 3

Time Left: 84 sec(s)

The _____ keyword specifies that you are creating a new trigger.

Select the correct option

☐

CAUSE

☒

CREATE☐☐

Reload Math Equations

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 83 sec(s)

The act of executing a trigger is also known as _____.

Select the correct option

 Reload Math Equations

☐	Finding a trigger
☐	Generate a trigger
☒	Firing a trigger
☐	Initiate a trigger

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 87 sec(s)

A/an _____ is a named PL/SQL block stored in the Oracle Database and executed automatically when a triggering event takes place.

Select the correct option

 Reload Math Equations

☒	Trigger
☐	Event
☐	Control block
☐	Index block

Click to Save Answer & Move to Next Question

Trigger's Disable/ Enable

Sometimes, you may want to disable a trigger for testing and troubleshooting purposes. To disable a trigger, you use the ALTER TRIGGER DISABLE statement:

```
ALTER TRIGGER trigger_name DISABLE;
```

```
ALTER TABLE table_name DISABLE ALL TRIGGERS;
```

Which of the following is correct statement to disable a trigger in PL/SQL?

Select the correct option

[Reload Math Equations](#)

☐	ALTER TRIGGER DISABLE;
☐	TRIGGER DISABLE;
☐	TRIGGER trigger_name DISABLE;
☒	ALTER TRIGGER trigger_name DISABLE;

[Click to Save Answer & Move to Next Question](#)

What is a Cursor?

- A cursor is a pointer that points to a result of a query.
- It is mainly used with tables which return more than one row.
- It is handled implicitly and explicitly.

Implicit Cursor

- Whenever Oracle executes an SQL statement such as SELECT INTO, INSERT, UPDATE, and DELETE, it automatically creates an implicit cursor.
- Oracle internally manages the whole execution cycle of implicit cursors and reveals only the cursor's information and statuses such as SQL%ROWCOUNT, SQL%ISOPEN, SQL%FOUND, and SQL%NOTFOUND.

Implicit Cursor

- The implicit cursor is not elegant when the query returns zero or multiple rows which cause NO_DATA_FOUND or TOO_MANY_ROWS exception respectively.

Explicit Cursor

- An explicit cursor is an SELECT statement declared explicitly in the declaration section of the current block or a package specification.
- For an explicit cursor, you have control over its execution cycle from OPEN, FETCH, and CLOSE.
- Oracle defines an execution cycle that executes an SQL statement and associates a cursor with it.

Execution Cycle of Explicit Cursor

Cursor's usage

- **Declare a cursor**

CURSOR cursor_name IS query;

- **Open a cursor**

OPEN cursor_name;

Cursor's usage

- **Fetch from a cursor**

The FETCH statement places the

contents of the current row into var.

FETCH cursor_name INTO variable_list;

To retrieve all rows in a result set, you

need to fetch each row till the last one.

- **Closing a cursor**

CLOSE cursor_name;

Explicit Cursor Attributes

- %ISOPEN
- %FOUND
- %NOTFOUND
- %ROWCOUNT

Implicit Cursor – Record Type

- A record is a composite data structure, which means that it is composed of one or more elements.
- Records are very much like a row of a database table, but each element of the record does not stand on its own. PL/SQL supports three kinds of records: table-based, cursor-based, and programmer-defined.

Which of the following is not an attribute of Explicit Cursor?

Select the correct option

Reload Math Equations

- ☐ %NOTFOUND
- ☒ %CLOSE
- ☐ %NOTFOUND
- ☐ %FOUND

Click to Save Answer & Move to Next Question

An explicit cursor is an statement declared explicitly in the declaration section of the current block or a package specification.

Select the correct option

Reload Math Equations

- ☐ REPLY
- ☐ INSERT
- ☐ REPLACE
- ☒ SELECT

Click to Save Answer & Move to Next Question

..... is mainly used with tables which return more than one row.

Select the correct option

Reload Math Equations

- ☒ Cursor
- ☐ Index
- ☐ Table
- ☐ Column

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 78 sec(s)

Which option is true about the following Code?
CURSOR c_emp IS
SELECT empno, ename, job
FROM emp
WHERE deptno=20;

Select the correct option

Reload Math Equations

☐	Cursor Retrieval
☐	Cursor Updation
☒	Cursor Declaration
☐	Cursor Deletion

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 73 sec(s)

Which option is true about the following Code?
DECLARE
CURSOR c_emp (p_job IN EMPJOB%TYPE) IS
SELECT *
FROM EMP WHERE job =p_job;
BEGIN
FOR r_emp IN c_emp('CLERK')
LOOP
DBMS_OUTPUT.PUT_LINE(r_emp.ename||'--'||r_emp.job);
END LOOP;
END;

Select the correct option

Reload Math Equations

☐	Cursor with Behavior
☐	Cursor with Pointers
☒	Cursor with Parameters
☐	Cursor with Transition Graphs

Question # 3 of 3

Time Left: 86 sec(s)

We may use cursor with _____

Select the correct option

Reload Math Equations

☐	Transition Graphs
☒	Parameters
☐	Pointers
☐	Behavior

Click to Save Answer & Move to Next Question

What is an Array?

All VARRAYS consist of contiguous memory locations. The lowest address corresponds to the first element and the highest address to the last element.

Declaration of an Array

A varray type is created with the **CREATE TYPE** statement. You must specify the maximum size and the type of elements stored in the varray.

CREATE Or REPLACE TYPE namearray AS VARRAY(3) OF VARCHAR2(10);

Question # 1 of 3

Time Left: 84 sec(s)

The lowest address corresponds to the _____ element and the highest address to the _____ element.

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|-------------------|
| ☐ | last, First |
| ☒ | First, last |
| ☐ | Last, Second Last |
| ☐ | First, Second |

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 82 sec(s)

The key in Associative Array can be either an integer or a_____.

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|---------|
| ☐ | Double |
| ☒ | String |
| ☐ | Boolean |
| ☐ | Float |

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 3

Time Left: 79 sec(s)

We must specify the _____ and the type of elements stored in the varray.

Select the correct option

 Reload Math Equations

☐	Number of arguments
☐	Type of arguments
☐	Datatype
☒	Maximum size

Click to Save Answer & Move to Next Question

_____ is a programming methodology for generating and running SQL statements at run time.

Select the correct option

 Reload Math Equations

☐	Light SQL
☒	Dynamic SQL
☐	Static SQL
☐	Static and Dynamic SQL

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 81 sec(s)

Which of the following statement is not true about Dynamic SQL?

Select the correct option

 Reload Math Equations

☐	It is useful when writing programs that must run database definition language (DDL) statements.
☐	It is useful when the number or data types of its input and output variables are known.
☐	It is useful when writing general-purpose and flexible programs like ad hoc query systems.
☒	It is useful when you know at compilation time the full text of a SQL statement.

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

Dynamic SQL is a programming methodology for generating and running SQL statements at

Select the correct option

[Reload Math Equations](#)

☐	None of these
☐	Compile time
☒	Run time
☐	Run and compile time

[Click to Save Answer & Move to Next Question](#)

Concepts of Oracle Instance-1

Introduction

A database instance is a set of memory structures that manage database files. A database is a set of physical files on disk created by the CREATE DATABASE statement. The instance manages its associated data and serves the users of the database.

Every running Oracle database is associated with at least one Oracle database instance. Because an instance exists in memory and a database exists on disk, an instance can exist without a database and a database can exist without an instance.

Configuration Option

During installation, oracle has an option to install instance, database or both

Database Instance Structure

When an instance is started, Oracle Database allocates a memory area called the system global area (SGA) and starts one or more background processes. The SGA serves various purposes, including the following:

- Maintaining internal data structures that are accessed by many processes and threads concurrently
- Caching data blocks read from disk
- Buffering redo data before writing it to the online redo log files
- Storing SQL execution plans

The SGA is shared by the Oracle processes, which include server processes and background processes, running on a single computer. The way in which Oracle processes are associated with the SGA varies according to operating system.

A database instance includes background processes. Server processes, and the process memory allocated in these processes, also exist in the instance. The instance continues to function when server processes terminate.

Question # 1 of 3

Time Left: 79 sec(s)

The system global area is shared by the Oracle processes, which include server processes and background processes, running on a _____ computer.

Select the correct option

 Reload Math Equations

☐	Different
☐	None of these
☐	Various
☒	Single

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 85 sec(s)

A Database is a set of physical files on disk created by the _____ statement.

Select the correct option

 Reload Math Equations

☐	INSERT DATABASE
☐	ADD DATABASE
☒	CREATE DATABASE
☐	GENERATE DATABASE

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

Every running Oracle database is associated with at least _____ Oracle database instance.

Select the correct option

 Reload Math Equations

☒	One
☐	Five
☐	Two
☐	Three

Click to Save Answer & Move to Next Question

Database Instance Configurations

You can run Oracle Database in either of the following mutually exclusive configurations:

■ Single-instance configuration

A one-to-one relationship exists between the database and an instance.

■ Oracle Real Application Clusters (Oracle RAC) configuration

A one-to-many relationship exists between the database and instances.

question # 1 of 3 Time Left: 81 sec(s)

In a **Single-instance configuration** A _____ relationship exists between the database and an instance.

Select the correct option Reload Math Equations

☒	one-to-one
☐	one-to-many
☐	many-to-many
☐	many-to-one

Click to Save Answer & Move to Next Question

In Oracle Real Application Clusters (Oracle RAC) configuration, A _____ relationship exists between the database and instances.

Select the correct option Reload Math Equations

☐	one-to-one
☒	one-to-many
☐	many-to-one
☐	many-to-many

Click to Save Answer & Move to Next Question

Question # 3 of 3 Time Left: 88 sec(s)

We can start a database instance and mount (associate the instance with) one database, but not mount _____ databases simultaneously with the same instance.

Select the correct option Reload Math Equations

☒	Two
☐	Four
☐	Five
☐	Three

Click to Save Answer & Move to Next Question

Duration of an Instance

An instance begins when it is created with the STARTUP command and ends when it is terminated. During this period, an instance can associate itself with one and only one database. Furthermore, the instance can mount a database only once, close it only once, and open it only once. After a database has been closed or shut down, you must start a *different* instance to mount and open this database.

Instance Startup

- Login with user sys / as sysdba
- After shutdown database, run the following:

Question # 1 of 3

Time Left: 88 sec(s)

The instance can mount a database only _____, close it only once, and open it only once.

Select the correct option

 Reload Math Equations

☐	Twice
☒	Once

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 86 sec(s)

The system identifier is _____ used by default to locate the _____

Select the correct option

 Reload Math Equations

☐	File directory
☒	Parameter file
☐	None of these
☐	Program file

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

An instance begins when it is created with the _____ command and ends when it is terminated.

Select the correct option

 Reload Math Equations

☒	STARTUP
☐	START
☐	BEGIN
☐	CREATE

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 87 sec(s)

A database _____ provides user access to a database.

Select the correct option

 Reload Math Equations

☒	Instance
☐	Row
☐	Table
☐	None of these

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 87 sec(s)

In a typical use case, which command can be used to start an instance and then mount and open the database, making it available for users.

Select the correct option

 Reload Math Equations

☐	CREATE
☐	BEGIN
☐	START
☒	STARTUP

Click to Save Answer & Move to Next Question

The data contained in the data files is accessible to _____.

Select the correct option

 Reload Math Equations

☐	Authorized and Unauthorized Users
☐	Unauthorized Users
☒	Authorized Users
☐	None of these

Click to Save Answer & Move to Next Question

Connection with Administrator Privileges

Database start-up and shutdown are powerful administrative options that are restricted to users who connect to Oracle Database with administrator privileges. Normal users do not have control over the current status of an Oracle database.

Depending on the operating system, one of the following conditions establishes administrator privileges for a user:

- The operating system privileges of the user enable him or her to connect using administrator privileges.
- The user is granted the SYSDBA or SYSOPER system privileges and the database uses password files to authenticate database administrators over the network.

When you connect with the SYSDBA system privilege, you are in the schema owned by SYS. When you connect as SYSOPER, you are in the public schema.

Database with Shutdown Instance

In a typical use case, you manually shut down the database, making it unavailable for users while you perform maintenance or other administrative tasks. You can use the SQL*Plus SHUTDOWN command or Enterprise Manager to perform these steps.

Progress from OPEN to Shutdown

Steps to perform Database Shutdown

Database closed

The database is mounted, but online data files and redo log files are closed.

Database unmounted

The instance is started, but is no longer associated with the control file of the database.

Database instance shut down

The database instance is no longer started.

Oracle Database does not go through all of the preceding steps in an instance failure or SHUTDOWN ABORT, which immediately terminates the instance.

SHUTDOWN ABORT

This mode is intended for emergency situations, such as when no other form of shutdown is successful. This mode of shutdown is the fastest. However, a subsequent open of this database may take substantially longer because instance recovery must be performed to make the data files consistent.

SHUTDOWN IMMEDIATE

This mode is typically the fastest next to SHUTDOWN ABORT. Oracle Database terminates any executing SQL statements and disconnects users. Active transactions are terminated and uncommitted changes are rolled back.

SHUTDOWN TRANSACTIONAL

This mode prevents users from starting new transactions, but waits for all current transactions to complete before shutting down. This mode can take a significant amount of time depending on the nature of the current transactions.

SHUTDOWN NORMAL

This is the default mode of shutdown. The database waits for all connected users to disconnect before shutting down.

Shutdown Modes Summary

A database administrator with SYSDBA or SYSOPER privileges can shut down the database using the SQL*Plus SHUTDOWN command or Enterprise Manager.

Question # 1 of 2

Time Left: 83 sec(s)

Temporary tablespaces created with the

Select the correct option

 Reload Math Equations

☐	INSERT TEMPORARY TABLESPACE
☐	NEW TEMPORARY TABLESPACE
☐	ADD TEMPORARY TABLESPACE
☒	CREATE TEMPORARY TABLESPACE

Click to Save Answer & Move to Next Question

Question # 2 of 2

Time Left: 82 sec(s)

..... users do not have control over the current status of an Oracle database.

Select the correct option

 Reload Math Equations

☒	Normal
☐	Privilege
☐	All
☐	Special

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 79 sec(s)

Active transactions are terminated and uncommitted changes are rolled back in

Select the correct option

 Reload Math Equations

☐	Start
☐	Shutdown
☐	Restart
☒	Immediate Shutdown

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 83 sec(s)

_____ mode prevents users from starting new transactions but waits for all current transactions to complete before shutting down.

Select the correct option

Reload Math Equations

☐	Immediate Shutdown
☐	Restart
☒	Shutdown Transactional
☐	Shutdown

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 82 sec(s)

Which of the following command is used to shut down the database, making it unavailable for users while you perform maintenance or other administrative tasks?

Select the correct option

Reload Math Equations

☐	mysql*Mul SHUTDOWN
☐	SQL*Div SHUTDOWN
☒	SQL*Plus SHUTDOWN
☐	All of these

Click to Save Answer & Move to Next Question

What is a Checkpoint?

A checkpoint is a crucial mechanism in consistent database shutdowns, instance recovery, and Oracle Database operation generally. The term checkpoint has the following related meanings:

- A data structure that indicates the checkpoint position, which is the SCN in the redo stream where instance recovery must begin

The checkpoint position is determined by the oldest dirty buffer in the database buffer cache. The checkpoint position acts as a pointer to the redo stream and is stored in the control file and in each data file header.

- The writing of modified database buffers in the database buffer cache to disk.

Whenever checkpoint happens all the dirty buffers of DB buffer cache (volatile storage) moves into data files (Non volatile storage).

Purpose of Checkpoints

Oracle Database uses checkpoints to achieve the following goals:

- Reduce the time required for recovery in case of an instance or media failure
- Ensure that dirty buffers in the buffer cache are written to disk regularly
- Ensure that all committed data is written to disk during a consistent shutdown

Recovery without Checkpoints

Extra time is wasted in processing the entire log file

Redoing of unnecessary transactions

Question # 1 of 3

Time Left: 85 sec(s)

Checkpoint is a mechanism in which all the previous logs are removed from the system and are stored permanently in the _____

Select the correct option

[Reload Math Equations](#)

☒	Storage disk
☐	None of these
☐	Buffer
☐	Cache

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 84 sec(s)

Which of the following is not the goal of Oracle Database when uses checkpoints :

Select the correct option

[Reload Math Equations](#)

☐	Ensure that all committed data is written to disk during a consistent shutdown
☐	Ensure that dirty buffers in the buffer cache are written to disk regularly
☒	Increase the time required for recovery incase of an instance or media failure
☐	Establish data consistency

[Click to Save Answer & Move to Next Question](#)

The checkpoint position acts as a _____ to the redo stream and is stored in the control file and in each data file header.

Select the correct option

 Reload Math Equations

☐	Heap
☒	Pointer
☐	Stack
☐	Array

Click to Save Answer & Move to Next Question

When to initiate Checkpoint

The checkpoint process (CKPT) is responsible for writing checkpoints to the data file headers and control file. Checkpoints occur in a variety of situations

Checkpoint Process

The checkpoint process (CKPT) updates the control file and data file headers with checkpoint information and signals DBWn to write blocks to disk. Checkpoint information includes the checkpoint position, SCN, location in online redo log to begin recovery, and so on.

Thread checkpoints

The database writes to disk all buffers modified by redo in a specific thread before a certain target. The set of thread checkpoints on all instances in a database is a database checkpoint. Thread checkpoints occur in the following situations:

- Consistent database shutdown
- ALTER SYSTEM CHECKPOINT statement
- Online redo log switch
- ALTER DATABASE BEGIN BACKUP statement

Tablespace and data file checkpoints

The database writes to disk all buffers modified by redo before a specific target. A tablespace checkpoint is a set of data file checkpoints, one for each data file in the tablespace. These checkpoints occur in a variety of situations, including making a tablespace read-only or taking it offline normal, shrinking a data file, or executing ALTER TABLESPACE BEGIN BACKUP.

Incremental checkpoints

An incremental checkpoint is a type of thread checkpoint partly intended to avoid writing large numbers of blocks at online redo log switches. DBWn checks at least every three seconds to determine whether it

has work to do. When DBWn writes dirty buffers, it advances the checkpoint position, causing CKPT to write the checkpoint position to the control file, but not to the data file headers.

POST-ASSESSMENT

Question # 1 of 3

Time Left: 84 sec(s)

An incremental checkpoint is a type of _____ checkpoint.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------|
| ☐ | Online |
| ☐ | Subject |
| ☐ | Offline |
| ☒ | Thread |

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 87 sec(s)

Shortest form of the Checkpoint Process is_____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|------|
| ☒ | CKPT |
| ☐ | CHPT |
| ☐ | CHPS |
| ☐ | CKPS |

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 71 sec(s)

Which of the following is not a checkpoint situation when it is occurs?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------------------------------|
| ☒ | Offline redo log switch |
| ☐ | ALTER DATABASE BEGIN BACKUP statement |
| ☐ | ALTER SYSTEM CHECKPOINT statement |
| ☐ | Consistent database shutdown |

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 87 sec(s)

Checkpoint command will run through _____.

Select the correct option

 Reload Math Equations

☐	User name
☐	User
☐	DBA
☒	Sys user

Click to Save Answer & Move to Next Question

At each switch of the redo log files _____ command will occur.

Select the correct option

 Reload Math Equations

☐	System privileges
☐	Alter
☒	Checkpoint
☐	Update

Click to Save Answer & Move to Next Question

Which option is true about the following statement?

Alter system checkpoint;

Select the correct option

 Reload Math Equations

☒	Checkpoint Occurring Command
☐	Checkpoint Updating Command
☐	Checkpoint Altering Command
☐	Checkpoint Deleting Command

Click to Save Answer & Move to Next Question

Overview of Instance Recovery

Instance recovery is the process of applying records in the online redo log to data files to reconstruct changes made after the most recent checkpoint. Instance recovery occurs automatically when an administrator attempts to open a database that was previously shut down inconsistently.

Question # 1 of 3

Time Left: 88 sec(s)

Instance recovery occurs _____ when an administrator attempts to open a database that was previously shut down inconsistently.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------|
| ☒ | Automatically |
| ☐ | None of these |
| ☐ | Manually |
| ☐ | Mechanically |

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 85 sec(s)

Instance recovery ensures that the database is in a _____ state after an instance failure.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--------------|
| ☐ | Inconsistent |
| ☒ | Consistent |

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

When an instance of an ORACLE database fails then the changes made by transaction is written _____.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|----------------------------|
| ☐ | Only in data file |
| ☒ | Only in redo log file |
| ☐ | Only in control file |
| ☐ | In redo log and data files |

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 84 sec(s)

A redo thread is marked _____ when the instance is shut down consistently.

Select the correct option

[Reload Math Equations](#)

☐	Shutdown
☒	Closed
☐	Opens
☐	None of these

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 83 sec(s)

Whether instance recovery is required depends on the state of the _____ threads.

Select the correct option

[Reload Math Equations](#)

☒	Redo
☐	Undo

Question # 3 of 3

Time Left: 88 sec(s)

A redo thread is marked open in the control file when a database instance _____ in read/write mode.

Select the correct option

[Reload Math Equations](#)

☐	Shutdown
☐	None of these
☒	Opens
☐	Closed

[Click to Save Answer & Move to Next Question](#)

Restore & Recovery Overview

To recover a SQL Server database from a failure, a database administrator has to restore a set of SQL Server backups in a logically correct and meaningful restore sequence.:

- The database (a complete database restore)
- The data file (a file restore)
- The data page (Bulk load, restore individual DB)

Overview of Restore Scenarios

A *restore scenario* in SQL Server is the process of restoring data from one or more backups and then recovering the database. The supported restore scenarios depend on the recovery model of the database and the edition of SQL Server.

- Complete DB Restore
- File Restore (individual files)

Steps to Restore a Database

To perform a database restore, the Database Engine executes three steps:

- Creates the database and transaction log files if they do not already exist.
- Copies all the data, log, and index pages from the backup media of a database to the database files.
- Applies the transaction log in what is known as the [recovery process](#).

Accelerated Database Recovery

- Accelerated Database Recovery (ADR) improves database availability, especially in the presence of long running transactions, by redesigning the SQL database engine recovery process. ADR is new in SQL Server 2019.
- ADR is available for Azure SQL Database, Azure SQL Managed Instance and Azure Synapse SQL.

Steps to Restore a Database

To perform a file restore, the Database Engine executes two steps:

- Creates any missing database file(s).
- Copies the data from the backup devices to the database file(s).

If you restore a file, you cannot recover it and bring it online until it has been rolled far enough forward to be consistent with the database.

Question # 1 of 3

Time Left: 78 sec(s)

Which of the following does not represent a database restore process?

Select the correct option

☐

Creates the database and transaction log files if they do not already exist.

☒

Copies the data from the backup devices to the database file(s).

☐

Applies the transaction log in what is known as the recovery process.

☐

Copies all the data, log, and index pages from the backup media of a database to the database files.

Reload Math Equations

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 82 sec(s)

ADR stand for

Select the correct option

 Reload Math Equations

☐	None of these
☒	Accelerated Database Recovery
☐	Accelerated Data Recovery
☐	Accelerated Recovery

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 55 sec(s)

A in SQL Server is the process of restoring data from one or more backups and then recovering the database.

Select the correct option

 Reload Math Equations

☐	Undo Phase
☐	None of these
☒	Restore Scenario
☐	Redo Phase

Click to Save Answer & Move to Next Question

Instance Recovery

Introduction

The first phase of instance recovery is called cache recovery or rolling forward, and involves reapplying all of the changes recorded in the online redo log to the data files.

Because rollback data is recorded in the online redo log, rolling forward also regenerates the corresponding undo segments.

Rolling Forward

Rolling forward proceeds through as many online redo log files as necessary to bring the database forward in time.

After rolling forward, the data blocks contain all committed changes recorded in the online redo log files. These files could also contain uncommitted changes that were either saved to the data files before the failure, or were recorded in the online redo log and introduced during cache recovery.

After Roll Forward

After the roll forward, any changes that were not committed must be undone. Oracle Database uses the checkpoint position, which guarantees that every committed change with an SCN lower than the checkpoint SCN is saved on disk. Oracle Database applies undo blocks to roll back uncommitted changes in data blocks that were written before

the failure or introduced during cache recovery. This phase is called rolling back or transaction recovery.

Question # 1 of 3

Time Left: 82 sec(s)

An alternate name of cache recovery is _____

Select the correct option

[Reload Math Equations](#)

- ☐ Segments
- ☐ Rollback
- ☐ None of these
- ☒ Rolling forward

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 84 sec(s)

The first phase of instance recovery is called _____

Select the correct option

[Reload Math Equations](#)

- ☒ Cache recovery
- ☐ Undo file
- ☐ Redo file
- ☐ None of these

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 3

Time Left: 88 sec(s)

Oracle Database can roll back _____ transactions simultaneously as needed.

Select the correct option

 Reload Math Equations

☐	Fewer
☐	Some
☐	None of these
☒	Multiple

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 82 sec(s)

Which of the following is not a basic variable parameter characteristic?

Select the correct option

 Reload Math Equations

☐	Database block size
☐	Locations of the control files
☐	Database name
☒	Redo tablespace

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 85 sec(s)

Purpose of variable parameters is to _____ database performance.

Select the correct option

 Reload Math Equations

☐	None of these
☐	Grow
☒	Improve
☐	Decrease

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 83 sec(s)

Oracle Database provides many initialization parameters to ----- its operation in diverse environments.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|----------|
| ☒ | Optimize |
| ☐ | Grow |
| ☐ | Double |
| ☐ | Multiple |

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 85 sec(s)

A server parameter file is a repository for initialization parameters that is managed by-----.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-----------------|
| ☐ | Database user |
| ☒ | Oracle Database |
| ☐ | System user |
| ☐ | DBA |

Question # 2 of 3

Time Left: 88 sec(s)

----- is a text file that contains a list of initialization parameters.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|------------------------------------|
| ☐ | Server parameter file |
| ☒ | Text initialization parameter file |

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

MEMORY_TARGET is an example of _____

Select the correct option

[Reload Math Equations](#)

☐	Session level parameters
☒	System-level parameters

[Click to Save Answer & Move to Next Question](#)

Overview of Diagnostic Files

Oracle Database includes a fault diagnosability infrastructure for preventing, detecting, diagnosing, and resolving database problems. Problems include critical errors such as code bugs, metadata corruption, and customer data corruption.

Goals of fault Diagnosability Infrastructure

- Detecting problems proactively
- Limiting damage and interruptions after a problem is detected
- Reducing problem diagnostic and resolution time
- Simplifying customer interaction with Oracle Support

Automatic Diagnostic Repository

Automatic Diagnostic Repository (ADR) is a file-based repository that stores database diagnostic data such as trace files, the alert log, and Health Monitor reports. Key characteristics of ADR include:

- Unified directory structure
- Consistent diagnostic data formats
- Unified tool set

Question # 1 of 3

Time Left: 87 sec(s)

Which of the following is the ADR root directory?

Select the correct option

[Reload Math Equations](#)

☐	None of these
☐	ADR dump
☒	ADR base
☐	ADR home

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 88 sec(s)

Which of the following is not a goals of the advanced fault diagnosability infrastructure?

Select the correct option

 Reload Math Equations

☐	Detecting problems proactively
☒	Increasing problem diagnostic and resolution time
☐	Simplifying customer interaction with Oracle Support
☐	Limiting damage and interruptions after a problem is detected

Question # 3 of 3

Time Left: 82 sec(s)

ADR is the abbreviation of?

Select the correct option

 Reload Math Equations

☐	Automated Diagnostic Repository
☐	Alternative Dispute Resolution
☒	Automatic Diagnostic Repository
☐	Alternative Diagnostic Repository

Click to Save Answer & Move to Next Question

Overview of Alert File

Each database has an alert log, which is an XML and text file containing a chronological log of database messages and errors. The alert log contents include the following:

- Deadlock occurs
- Any error written by Oracle Server process
- Any unexpected messages and corruptions happened
- Any DBA DDL operations are managed

Question # 1 of 3

Time Left: 70 sec(s)

----- file constantly be monitored to detect unexpected messages and corruptions in database.

Select the correct option

 Reload Math Equations

☐	Redo log
☒	Alert log
☐	None of these
☐	Undo log

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 88 sec(s)

----- is an XML file containing a chronological log of database messages and errors.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------|
| ☐ | Redo log |
| ☐ | Undo log |
| ☒ | Alert log |
| ☐ | None of these |

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

Which of the following is not a content of alert log file?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--|
| ☐ | Internal errors |
| ☐ | Block corruption error |
| ☒ | Default initialization parameter used at startup |
| ☐ | Non default initialization parameter used at startup |

Click to Save Answer & Move to Next Question

Overview of Trace Files

A **trace file** is an administrative file that contain diagnostic data used to investigate problems. Also, trace files can provide guidance for tuning applications or an instance.

Question # 1 of 3

Time Left: 88 sec(s)

----- is a special type of trace file.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------|
| ☐ | Alert |
| ☒ | Dump |
| ☐ | None of these |
| ☐ | Incident |

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 87 sec(s)

_____ files can be useful for tuning applications or instances.

Select the correct option

 Reload Math Equations

☐	Log files
☒	Trace files
☐	Redo log files
☐	None of these

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

A _____ is an administrative file that contain diagnostic data used to investigate problems.

Select the correct option

 Reload Math Equations

☐	Log file
☒	Trace file
☐	Redo log file
☐	None of these

Click to Save Answer & Move to Next Question

Reading Material:

Types of Security

Following are the database security types and issues

- Legal & Ethical Issues
- Policy Issues in Enterprises
- Systems' related Issues
- Needs of Enterprises

Threats to Databases

Following are the database related threats and issues

- Loss of integrity
- Loss of availability
- Loss of Confidentiality

Security Domain

The database administrator defines the names of the users who are allowed to access a database. A security domain defines the settings that apply to the user.

Authentication mechanisms are

- Data Dictionary
- Operating System
- Network

Schema Objects & User

- A Schema is a collection of objects
- A user is associated (owner) of schema
- Creating schema means creating a user as an owner of schema

User & Database

- Accessing Database (connection with the database)
- Specifying database unique user account of ID
- Specifying authentication (roles & privileges) to user account

Allocating space, quota, buffer etc. are additional features in security.

Creating Oracle User

- A unique user name or account
- Authentications (roles/ rights)
- A default tablespace
- A temporary tablespace
- A user profile
- An account status (lock/unlock)

Pre-defined Accounts

- SYS account (sys as sysdba)
 - is granted the DBA role
 - has all privileges with ADMIN OPTION
 - is used for startup, shutdown etc.
 - owns the data dictionary
- SYSTEM account is granted the DBA role
 - Normally not used for routine operations

Question # 1 of 3

Time Left: 87 sec(s)

Schema is a collection of _____.

Select the correct option

 Reload Math Equations

- ☐ Functions
- ☐ Synonyms
- ☒ Objects
- ☐ Procedures

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 88 sec(s)

Which of the following option is not related to database security types and its issues?

Select the correct option

 Reload Math Equations

- ☐ Needs of Enterprises
- ☐ Legal & Ethical Issues
- ☒ Loss of integrity
- ☐ Policy Issues in Enterprises

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

The names of the users who are allowed to access a database are defined by the _____.

Select the correct option

 Reload Math Equations

- ☐ None of these
- ☒ Database administrator
- ☐ User
- ☐ System administrator

Click to Save Answer & Move to Next Question

Types of Oracle User Privileges

- System: Enables users to perform particular actions in the database
- Object: Enables users to access and manipulate a specific object

System Privileges

- More than 100 distinct system privileges
- ANY keyword in privileges signifies that users have the privilege in any schema
- GRANT command adds a privilege to a user or a group of users
- REVOKE command deletes the privileges

Question # 1 of 3

Time Left: 84 sec(s)

----- privileges enable users to access and manipulate a specific object.

Select the correct option

[Reload Math Equations](#)

☐	System
☒	Object

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 84 sec(s)

----- privileges enable users to perform particular actions in the database.

Select the correct option

[Reload Math Equations](#)

☒	System
☐	Object

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 3

Time Left: 85 sec(s)

Which of the following command is used to delete the system privileges from oracle?

Select the correct option

[Reload Math Equations](#)

☐	Delete
☐	None of these
☒	Revoke
☐	Grant

[Click to Save Answer & Move to Next Question](#)

What is User Privileges?

A user privilege is the right to run a particular type of SQL statement, or the right to access an object that belongs to another user, run a PL/SQL package, and so on.

- System Privileges
- Object Privileges
- Roles

The screenshot shows a quiz interface with a question box, a list of options, and navigation buttons. The question is: "Which privileges is used to run a particular type of SQL statement, or the right to access an object that belongs to another user?". The options are: Object Privileges, None of these, System Privileges, and User Privileges. The "User Privileges" option is selected. There are buttons for "Reload Math Equations" and "Click to Save Answer & Move to Next Question".

Question # 1 of 1 Time Left: 86 sec(s)

Which privileges is used to run a particular type of SQL statement, or the right to access an object that belongs to another user?

Select the correct option Reload Math Equations

- ☐ Object Privileges
- ☐ None of these
- ☐ System Privileges
- ☒ User Privileges

Click to Save Answer & Move to Next Question

Managing Roles

Introduction

- Roles are named groups of related privileges or rights.
- Each user can be granted roles and privileges for each role, giving them a tailored set of abilities.
- Roles define what actions the user can take, and privileges specify the targets to which their roles apply.
- You can view the permissions granted by each role, add and remove users, and assign roles and notification profiles to users.

Data Dictionary for Privileges

We can use string search PRIV to display possible data dictionary views.

For Example,

```
select table_name
```

```
from dict
```

```
where table_name like '%PRIV%';
```

Post-Assessment

Question # 1 of 3Time Left: 84 sec(s)

Which privileges is used to run a particular type of SQL statement, or the right to access an object that belongs to another user?

Select the correct option

☒

User Privileges

☐

None of these

☐

Object Privileges

☐

System Privileges

Click to Save Answer & Move to Next Question

Reload Math Equations

Question # 2 of 3Time Left: 85 sec(s)

Which of the following command is used to give the privileges to users?

Select the correct option

☐

Revoke

☐

Create

☐

Delete

☒

Grant

Click to Save Answer & Move to Next Question

Reload Math Equations

Question # 3 of 3Time Left: 83 sec(s)

In Oracle, which string search is used to display the possible data dictionary views?

Select the correct option

☐

DICT

☒

PRIV

Click to Save Answer & Move to Next Question

Reload Math Equations

_____ is a system defined privilege that control access to a specific object.

Select the correct option

 Reload Math Equations

☐	User Privilege
☒	Object Privilege
☐	System Privilege
☐	Roles

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 87 sec(s)

_____ are named groups of related privileges or rights.

Select the correct option

 Reload Math Equations

☐	Object Privileges
☐	User Privileges
☐	System Privileges
☒	Roles

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 86 sec(s)

_____ privileges allow users to perform specific database operations.

Select the correct option

 Reload Math Equations

☐	None of these
☐	Object
☒	System
☐	User

Click to Save Answer & Move to Next Question

Benefits of Roles

- Easier privilege management
- Dynamic privilege management
- Selective availability of privileges
- Can be granted through the operating system

Easier privilege management

Use roles to simplify privilege management. Rather than granting the same set of privileges to several users, you can grant the privileges to a role, and then grant that role to each user.

Dynamic privilege management

If the privileges associated with a role are modified, all the users who are granted the role acquire the modified privileges automatically and immediately.

Selective availability of privileges

Roles can be enabled or disabled to turn privileges on or off temporarily. Enabling a role can also be used to verify that a user has been granted that role.

Granting through the operating system

Operating system commands or utilities can be used to assign roles to users in the database.

Standard and Default Roles

- CONNECT is a role that Enterprise Manager automatically grants to a user when you create a user as shown in "Creating Users". This role has the CREATE SESSION privilege.
- RESOURCE extends the privileges of a user beyond those granted by the CONNECT role. It includes CREATE PROCEDURE, CREATE TRIGGER, and other system privileges.
- DBA is the standard role that can be granted by an administrator to another administrator. It includes all system privileges and should only be granted to the most trusted and qualified of users. Assigning this role to a user enables the user to administer the database.

Roles of User SYS

- When you create an Oracle database, the user SYS is automatically created and granted the DBA role.
- All base tables and views for the database data dictionary are stored in the schema SYS. These base tables and views are critical for the operation of Oracle Database.
- To maintain the integrity of the data dictionary, tables in the SYS schema are manipulated only by the database. They should never be modified by any user or database administrator.
- Also, you should not create any tables in the schema of user SYS, although you can change the storage parameters of the data dictionary settings if necessary.

Analyzing Roles & Privileges to Users

- Assuming there are three departments in some Enterprise
 - Sales
 - Accounting
 - Human Development & Resource (HR)
- Employees are working as an external users in these departments
- Analyzing how Roles and Privileges are need to be decided
- Users Ali(Sales), Jamil(HR) and Nadeem (Accounting)

Characteristics of Roles

- Roles can be granted to and revoked from users with the same commands that are used to grant and revoke system privileges.
- Roles can be granted to any user or role. However, a role cannot be granted to itself and cannot be granted circularly.
- A role can consist of both system and object privileges.
- A role can be enabled or disabled for each user who is granted the role.
- A role can be required a password to be enabled.
- Each role name must be unique among existing usernames and role names.
- Roles are not owned by anyone; and they are not in any schema.
- Roles have their description stored in the data dictionary.

Question # 1 of 4

Time Left: 62 sec(s)

Which of the following options does not fall within the category of role benefits?

Select the correct option

 Reload Math Equations

☐	Easier privilege management
☐	Selective availability of privileges
☒	Resource Privileges
☐	Dynamic privilege management

Click to Save Answer & Move to Next Question

Question # 2 of 4

Time Left: 85 sec(s)

Roles can be granted to **any** user.

Select the correct option

 Reload Math Equations

☐	Special
☒	Any
☐	All of these
☐	Specific

Click to Save Answer & Move to Next Question

Question # 3 of 4

Time Left: 74 sec(s)

A role can be required a _____ to be enabled.

Select the correct option

Reload Math Equations

☐	Username
☒	Password

Click to Save Answer & Move to Next Question

Question # 4 of 4

Time Left: 63 sec(s)

_____ extends the privileges of a user beyond those granted by the CONNECT role.

Select the correct option

Reload Math Equations

☐	CONNECT
☐	DBA
☒	RESOURCE
☐	None of these

Click to Save Answer & Move to Next Question

Reading Material:

SQL-Create Role

- Use the CREATE ROLE statement to create roles. You must have the CREATE ROLE system privilege to create roles. When you create a role that is not identified externally or by password, the role is granted with the ADMIN option.
- You need to decide set of privileges to put in one role and a user who will use those privileges by assigning just role.

Syntax of SQL-Create Role

Use the following command to create a role:

CREATE ROLE role [NOT IDENTIFIED | IDENTIFIED {BY password | EXTERNALLY | GLOBALLY | USING package}]

where:

- role: Is the name of the role

- NOT IDENTIFIED: Indicates that no verification is required when enabling the role
- IDENTIFIED: Indicates that verification is required when enabling the role
- BY password: Provides the password that the user must specify when enabling the role
- USING package: Creates an application role, which is a role that can be enabled only by applications using an authorized package
- EXTERNALLY: Indicates that a user must be authorized by an external service (such as the operating system or a third-party service) before enabling the role
- GLOBALLY: Indicates that a user must be authorized to use the role by the enterprise directory service before the role is enabled with the SET ROLE statement, or at login.

Roles with ADMIN option

- Not identified:

Create role oe_clerk;
- By password:

Create role hr_clerk

IDENTIFIED BY bonus;
- Identified externally:

Create role hr_manager

identified Externally;

Syntax of SQL-Create Role using EM

Role Name	Description
CONNECT, RESOURCE, DBA	These roles are provided for backward compatibility
EXP_FULL_DATABASE	Privileges to export the database
IMP_FULL_DATABASE	Privileges to import the database
DELETE_CATALOG_ROLE	DELETE privileges on data dictionary tables
EXECUTE_CATALOG_ROLE	EXECUTE privilege on data dictionary packages
SELECT_CATALOG_ROLE	SELECT privilege on data dictionary tables

Question # 2 of 2

Time Left: 86 sec(s)

_____ clause indicates that verification is required when enabling the role

Select the correct option

 Reload Math Equations

☐	NOT IDENTIFIED
☒	IDENTIFIED
☐	BY password
☐	GLOBALLY

[Click to Save Answer & Move to Next Question](#)

Alter Role

- ALTER ROLE modifies the authentication method.
- Modifying roles requires the ADMIN option or ALTER ANY ROLE privilege

Default Role

A user can be assigned many roles. A default role is a subset of these roles that is automatically enabled when the user logs on. By default, all the roles assigned to a user are enabled at logon without the need of a password. Limit the default roles for a user with the *ALTER USER* command.

----- command is used to limit the default roles for a user.

Select the correct option

Reload Math Equations

☐ CREATE ROLE

☐ ALTER ROLE

☐ DEFAULT ROLE

☒ ALTER USER

Click to Save Answer & Move to Next Question

Question # 2 of 3

Time Left: 88 sec(s)

----- role modifies the authentication method.

Select the correct option

Reload Math Equations

☒ ALTER

☐ CREATE

☐ EXCEPT

☐ DEFAULT

Click to Save Answer & Move to Next Question

Question # 1 of 3

Time Left: 84 sec(s)

----- clause indicates that a user must be authorized to use the role by the enterprise directory service before the role is enabled with the SET ROLE statement, or at login.

Select the correct option

Reload Math Equations

☒ GLOBALLY

☐ EXTERNALLY

☐ NOT IDENTIFIED

☐ IDENTIFIED

Click to Save Answer & Move to Next Question

Application Roles

- Application roles can be enabled only by authorized PL/SQL packages
- The USING package clause creates an application role

Enable & Disable Role

- Disable a role to temporarily revoke the role from a user.

- Enable a role to grant it temporarily.
- The SET ROLE command enables and disables roles
- Default roles are enabled for a user at login.
- A password may be required to enable a role.
- Enable and disable roles to temporarily activate and deactivate the privileges associated with the roles.
- To enable a role, a role must first be granted to the user.
- When a role is enabled, the user can use the privileges granted to that role. If a role is disabled, the user cannot use the privileges associated with that role unless those privileges are granted directly to a user or to another role enabled for that user. Roles are enabled for a session. At the next session, the user's active roles revert to default roles.

Specifying Roles to be Enabled

- The SET command and the DBMS_SESSION.SET_ROLE procedure enable all of the roles included in the command and disable all other roles. Roles can be enabled from any tool or program that allows PL/SQL commands; however, a role cannot be enabled in a stored procedure.
- You can use the ALTER USER-DEFAULT ROLE command to indicate which roles will be enabled for a user at login. All other roles are disabled.
- A password may be required to enable a role. The password must be included in the SET ROLE command to enable the role.
- Default roles assigned to a user do not require a password; they are enabled at login, the same as a role without a password.

Restrictions on Enable & Disable Role

- A role cannot be enabled from a stored procedure, because this action may change the security domain (set of privileges) that allowed the procedure to be called in the first place.
- So, in PL/SQL, Role can be enabled and disabled in anonymous blocks and application procedures (for example, client Oracle Forms procedures), but not in stored procedures.

Question # 1 of 3

Time Left: 86 sec(s)

----- command is used to indicate which roles will be enabled for a user at login.

Select the correct option

[Reload Math Equations](#)

☐	MODIFY USER-DEFAULT ROLE
☐	ALL OF THESE
☒	ALTER USER-DEFAULT ROLE
☐	CREATE USER-DEFAULT ROLE

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 3

Time Left: 87 sec(s)

----- clause is used to create an application role.

Select the correct option

 Reload Math Equations

☐	Globally
☐	Externally
☒	Using
☐	Delete

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 88 sec(s)

----- command enables and disables roles.

Select the correct option

 Reload Math Equations

☐	MODIFY ROLE
☐	UPDATE ROLE
☒	SET ROLE
☐	CREATE ROLE

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 81 sec(s)

Application roles can be enabled only by ----- PL/SQL packages.

Select the correct option

 Reload Math Equations

☒	Authorized
☐	Unauthorized

Click to Save Answer & Move to Next Question

Reading Material:

Guidelines

Because a role includes the privileges that are necessary to perform a task, the role name is usually an application task or a job title.

Our example uses both application tasks and job titles for role names.

Guidelines for an Example

Use the following steps to create, assign, and grant users roles:

- Create a role for each application task. The name of the application role corresponds to a task in the application, such as PAYROLL.
- Assign the privileges necessary to perform the task to the application role.
- Create a role for each type of user. The name of the user role corresponds to a job title, such as PAY_CLERK.
- Grant application roles to user's roles.
- Grant user's roles to users.

If a modification to the application requires that the new privileges are needed to perform the payroll task, then the DBA only needs to assign the new privileges to the PAYROLL application role. All of the users that are currently performing this task will receive the new privileges.

Guidelines for Assigning Roles

Guidelines for using Password and Default Roles

Passwords provide an additional level of security when enabling a role.

For example, the application might require a user to enter a password when enabling the PAY_CLERK role, because this role can be used to issue checks.

Passwords allow a role to be enabled only through an application. This technique is shown in the example in the slide.

- The DBA has granted the user two roles, PAY_CLERK and PAY_CLERK_RO.
- The PAY_CLERK has been granted all of the privileges that are necessary to perform the payroll clerk function.

Roles with Password

Question # 1 of 3

Time Left: 85 sec(s)

----- privileges can also be used to revoke any role from a user

Select the correct option

 Reload Math Equations

☐	ALTER ANY ROLE
☐	NONE OF THESE
☐	CREATE ANY ROLE
☒	GRANT ANY ROLE

Click to Save Answer & Move to Next Question

----- Which SQL statement is used to remove a role from the database?

Select the correct option

 Reload Math Equations

☒	DROP ROLE role;
☐	REMOVE ROLE role;
☐	DELETE ROLE role;
☐	ALTER ROLE role;

Click to Save Answer & Move to Next Question

Question # 3 of 3

Time Left: 86 sec(s)

----- provide an additional level of security when enabling a role

Select the correct option

 Reload Math Equations

☐	Username
☐	None of these
☒	Passwords
☐	PIN code

Click to Save Answer & Move to Next Question

Which of the following SQL statement is used to revoke a role from a user?

Select the correct option

 Reload Math Equations

☐	Create
☐	Alter
☐	All of these
☒	Revoke

Click to Save Answer & Move to Next Question